

Simple Linear Regression

$$\begin{aligned} J(w) &= \frac{1}{2} \sum_i^n (\hat{y}_i - y_i)^2 \\ &= \frac{1}{2} \sum_i^n (w_0 + w_1 x_i - y_i)^2 \end{aligned} \tag{1}$$

To find the weights that minimize the cost, compute partial derivatives:

$$\nabla J(w) = \begin{bmatrix} \frac{\partial J}{\partial w_0} \\ \frac{\partial J}{\partial w_1} \end{bmatrix} \tag{2}$$

Lets start by finding the w_0 :

$$\frac{\partial J}{\partial w_0} = \sum_i^n (w_0 + w_1 x_i - y_i) \tag{3}$$

Now set this to 0.

$$\begin{aligned} \sum_i^n (w_0 + w_1 x_i - y_i) &= 0 \\ \sum_i^n w_0 + w_1 \sum_i^n x_i - \sum_i^n y_i &= 0 \\ nw_0 + w_1 \sum_i^n x_i - \sum_i^n y_i &= 0 \\ nw_0 &= \sum_i^n y_i - w_1 \sum_i^n x_i \\ w_0 &= \frac{1}{n} \sum_i^n y_i - \frac{1}{n} w_1 \sum_i^n x_i \\ w_0 &= \mu_y - w_1 \mu_x \end{aligned} \tag{4}$$

μ_y is the average of all y values

Now we repeat the process for w_1 :

$$\begin{aligned}
\frac{\partial J}{\partial w_0} &= \sum_i^n (w_0 + w_1 x_i - y_i) x_i \\
0 &= \sum_i^n (w_0 + w_1 x_i - y_i) x_i \\
0 &= \sum_i^n (w_0 x_i + w_1 x_i^2 - y_i x_i) \\
0 &= w_0 \sum_i^n x_i + w_1 \sum_i^n x_i^2 - \sum_i^n y_i x_i \\
w_0 \sum_i^n x_i + w_1 \sum_i^n x_i^2 &= \sum_i^n y_i x_i \\
(\mu_y - w_1 \mu_x) \sum_i^n x_i + w_1 \sum_i^n x_i^2 &= \sum_i^n y_i x_i \\
\mu_y \sum_i^n x_i - w_1 \mu_x \sum_i^n x_i + w_1 \sum_i^n x_i^2 &= \sum_i^n y_i x_i \\
-w_1 \mu_x \sum_i^n x_i + w_1 \sum_i^n x_i^2 &= \sum_i^n y_i x_i - \mu_y \sum_i^n x_i \\
w_1 \left(\sum_i^n x_i^2 - \mu_x \sum_i^n x_i \right) &= \sum_i^n y_i x_i - \mu_y \sum_i^n x_i \\
w_1 &= \frac{\sum_i^n y_i x_i - \mu_y \sum_i^n x_i}{\left(\sum_i^n x_i^2 - \mu_x \sum_i^n x_i \right)}
\end{aligned} \tag{5}$$

Looking at the numerator:

$$\begin{aligned}
\sum_i^n y_i x_i - \mu_y \sum_i^n x_i &= \sum_i^n y_i x_i - n \mu_x \mu_y \\
&= \sum_i^n y_i x_i - n \mu_x \mu_y - n \mu_x \mu_y + n \mu_x \mu_y \\
&= \sum_i^n y_i x_i - \mu_y \sum_i^n x_i - x \sum_i^n y_i + \sum_i^n \mu_x \mu_y \\
&= \sum_i^n (x_i y_i - x_i \mu_y - \mu_x y_i + \mu_x \mu_y) \\
&= \sum_i^n (x_i - \mu_x)(y_i - \mu_y)
\end{aligned} \tag{6}$$

Looking at the denominator:

$$\begin{aligned}\sum_i^n x_i^2 - \mu_x \sum_i^n x_i &= \sum_i^n x_i^2 - n\mu_x^2 \\ &= \sum_i^n x_i^2 - 2n\mu_x\mu_x + n\mu_x^2 \\ &= \sum_i^n x_i^2 - 2\mu_x \sum_i^n x_i - \sum_i^n \mu_x^2 \\ &= \sum_i^n (x_i^2 - 2\mu_x x_i + \mu_x^2) \\ &= \sum_i^n (x_i - \mu_x)^2\end{aligned}\tag{7}$$

Thus:

$$w_1 = \frac{\frac{1}{n-1} \sum_i^n (x_i - \mu_x)(y_i - \mu_y)}{\frac{1}{n-1} \sum_i^n (x_i - \mu_x)^2}\tag{8}$$

Finally,

$$\nabla J(w) = \begin{bmatrix} \frac{\partial J}{\partial w_0} \\ \frac{\partial J}{\partial w_1} \end{bmatrix} = \begin{bmatrix} \mu_y - w_1\mu_x \\ \frac{\text{Cov}(x, y)}{\text{Var}(x)} \end{bmatrix}\tag{9}$$

This is called Ordinary Least Squares.

Normal Equations

Your task now is to take the cost function and derive the normal equation. Remember, take the derivative in regards to $\vec{w}$, set the derivative to 0 and then solve for w .

$$\begin{aligned} J(\vec{w}) &= \frac{1}{2}(wX - y)^2 \\ &= \frac{1}{2}(wX - y)^T(wX - y) \\ &= \frac{1}{2}(X^T w^T wX - X^T w^T y - y^T wX - y^T y) \\ &= \frac{1}{2}((wX)^2 - 2wX^T y - y^T y) \end{aligned} \tag{10}$$

Now take the derivative in terms of $\vec{w}$:

$$\frac{dJ(\vec{w})}{\partial w} = (wX)X - X^T y \tag{11}$$

Now set to 0 and solve for $\vec{w}$.

$$\begin{aligned} 0 &= (wX)X - X^T y \\ wX^T X &= X^T y \\ w &= (X^T X)^{-1} X^T y \end{aligned} \tag{12}$$

By definition, $(X^T X)$ will be a symmetric matrix. This also assumes that $(X^T X)$ is non-singular, i.e. the matrix can be inverted, the matrix is full rank, the columns and rows of the matrix are linearly independent.